UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PRE CALCULUS
COURSE CODE	MAT133
EXAMINATION	OCTOBER 2026
TIME	2 HOURS

NSTRUCTIONS TO CANDIDATES

1. This question paper consists of 5 questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) a one – page Appendix 1
   iii) an Answer Booklet - provided by the Faculty
   iv) a graph paper - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 7 printed pages

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QUESTION 1

a) Solve the inequality $-x^2 + 2x + 15 \geq 0$.

(5 marks)



Figure: critical values.

b) Solve the inequality $|x + 5| \leq 2$. Express the answer in interval notation and on a number line.

(5 marks)



Figure: solution on a number line.

c) Sketch $y = 2|x - 3| + 1$. State the vertex, the y-intercept and any x-intercepts.

(5 marks)

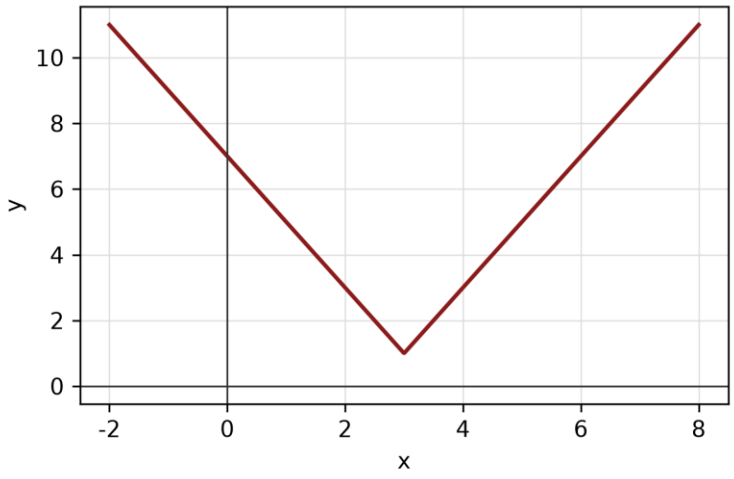


Figure: absolute-value graph.

QUESTION 2

a) Given $f(x) = (4x - 6)/(3x + 5)$, $x \neq -5/3$, find $f^{-1}(x)$ and state its domain.

(5 marks)

b) Apply graph-transformation techniques to sketch $y = -3(x - 5)^2 - 3$. State the vertex and the y-intercept.

(5 marks)

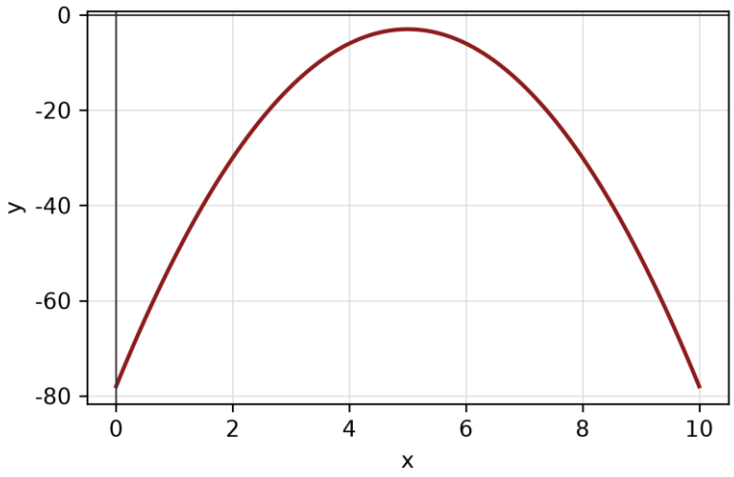


Figure: transformed graph.

c) Write $x^2 + 8x + y^2 - 4y - 5 = 0$ in standard form. Hence state the centre and radius and sketch the circle.

(5 marks)

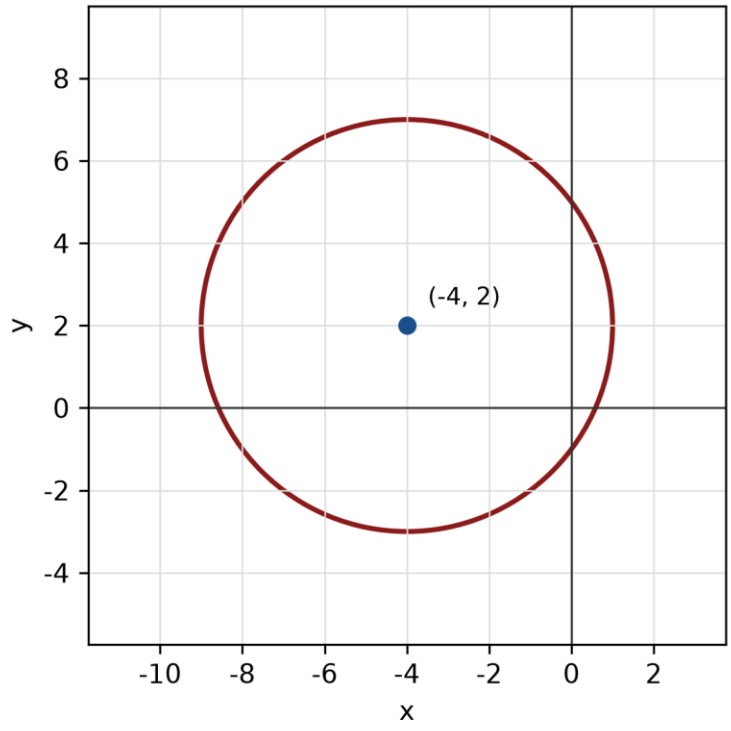


Figure: circle.

QUESTION 3

a) Given $z_1 = 4 - 3i$ and $z_2 = 3 - 5i$, find $z_1 + z_2$ in the form $a + bi$.

(4 marks)

b) Find the modulus and argument of $z = 3 + 4i$. Give the argument in degrees correct to one decimal place.

(5 marks)

c) Use the remainder theorem to find the remainder when $f(x) = 2x^3 + 5x^2 + 3x + 4$ is divided

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by x - 2.

(3 marks)

QUESTION 4

a) Solve the system of equations simultaneously:
-2x + y - 3z = 5
2x - y + 2z = -1
-x + y + 3z = -17

(5 marks)

b) Graph the system of inequalities below and shade the feasible region. State the vertices of the region.
x ≥ 0, y ≥ 0, 3x + 1y ≤ 5.

(5 marks)

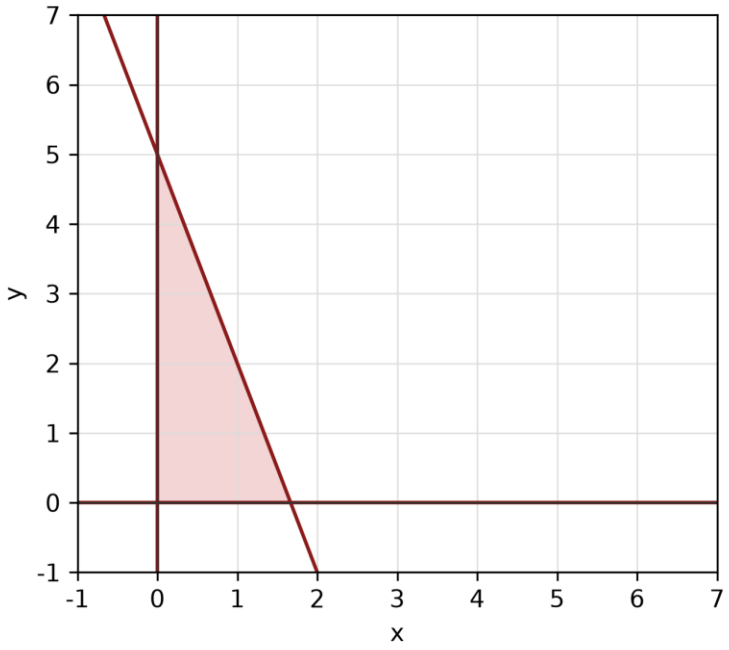


Figure: feasible region.

c) Find the equation of the straight line passing through (0, 0) and (2, -4).

(4 marks)

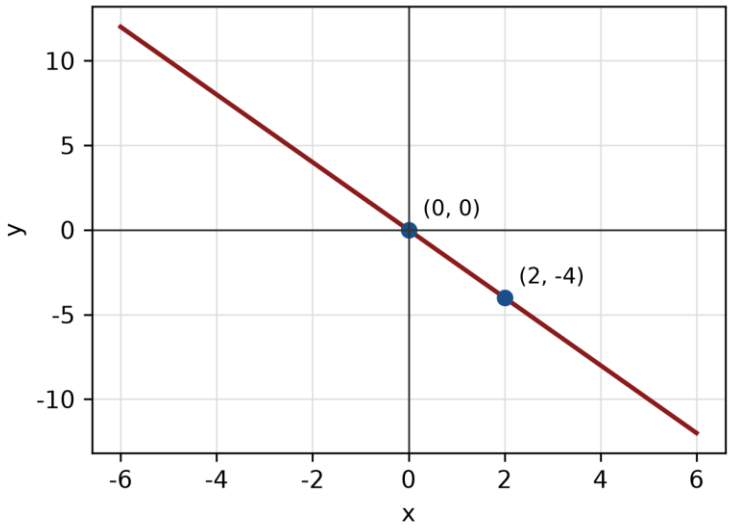


Figure: line through the two points.

QUESTION 5

a) Prove the identity $\sin^2 x/(1 - \cos x) = 1 + \cos x$.

(5 marks)

b) Find the perimeter of the sector with radius 11 cm and central angle 30°.

(4 marks)

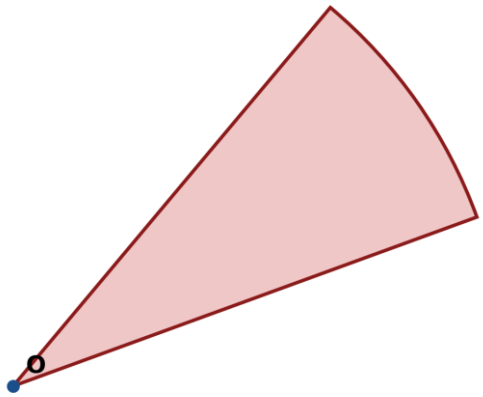


Figure: sector.

c) In triangle ABC, b = 11 cm, c = 7 cm and A = 90°. Use the cosine rule to find a.

(4 marks)

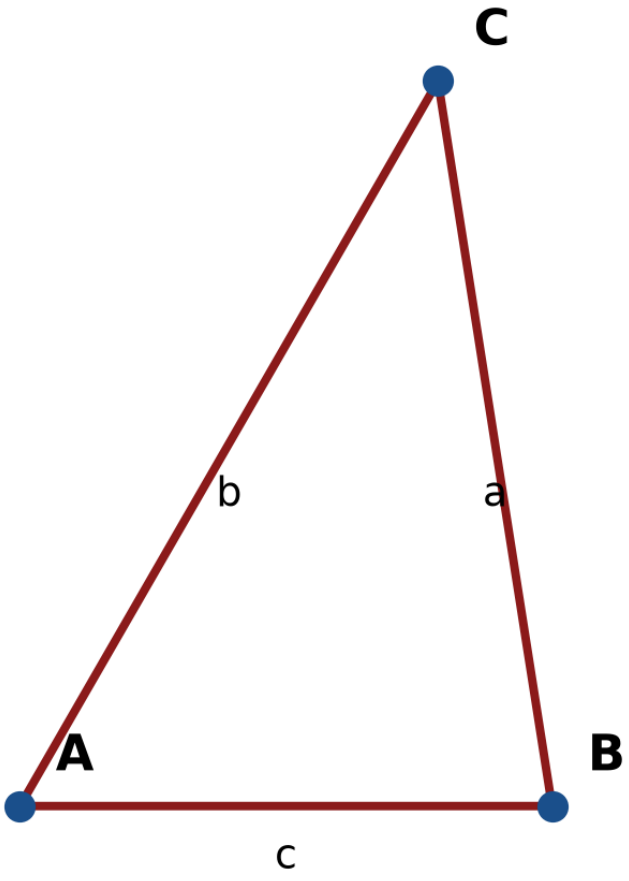


Figure: triangle ABC.

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END OF QUESTION PAPER

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APPENDIX 1

Formula sheet (as supplied with the paper).

$\sin^2 A + \cos^2 A = 1$

$\sin(A + B) = \sin A \cos B + \cos A \sin B$

$\cos(A + B) = \cos A \cos B - \sin A \sin B$

$\tan(A - B) = (\tan A - \tan B) / (1 + \tan A \tan B)$

$\cos 2A = \cos^2 A - \sin^2 A$

$\cos 2A = 1 - 2\sin^2 A = 2\cos^2 A - 1$

$\sin 2A = 2 \sin A \cos A$

$\tan^2 A + 1 = \sec^2 A$

$\tan 2A = 2 \tan A / (1 - \tan^2 A)$

Arc length, $s = r\theta$

Area of sector = $\frac{1}{2}r^2\theta$

$a / \sin A = b / \sin B = c / \sin C$

$a^2 = b^2 + c^2 - 2bc \cos A$

Area of triangle = $\frac{1}{2}ab \sin C$

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